Guoxiaoqing(Lucy) Sun

Links: [GitHub Personal Page] [LinkedIn]

SKILLS

Coding Language	C/C#/C++, Verilog/SystemVerilog, VHDL, Python, ARM Assembly
Embedded/Hardware	FPGA, STM32, ARM Cortex-M, RISC-V, UART, SPI, SERDES, RTL simulation.
Tools	Git, Vivado, Quartus, STM32CubeIDE, Linux, MATLAB, VS Code, MySQL
Language	English, Mandarin

WORK EXPERIENCE

Embedded Systems Intern, *Shenzhen Dahuan Robotics Technology Co Ltd* *Jan 2026 – Apr 2026*

- Implemented **FPGA-based** communication modules for SPI, UART/RS485, and SERDES interfaces, enabling board-level sensor communication between two chips in a planar motor-control FPGA system.
- Verified stable 20 MHz **SPI** communication across multi-sensor boards through debugging and long-duration testing.
- Developed **UART/RS485** logic for laser sensor polling with half-duplex control, frame parsing, and CRC checking to improve communication reliability.
- Debugged **SERDES** transmission with 8b/10b encoder/decoder, FIFO buffering, comma detection, and waveform capture.
- Documented communication debugging procedures and verification results to support future hardware development.

Research & Development Assistant, *Wizard Technology (Shanghai)* *May 2024 – Aug 2024*

- Built and tested simulation circuits in **Cadence Virtuoso** to support analog/mixed-signal layout verification workflows.
- Configured **Linux-based CAD platforms** and virtual machine environments, improving setup consistency for circuit design and simulation tasks.
- Processed image and audio datasets using **MATLAB** filters to support AI-assisted layout automation experiments.

Senior Process Mechanical Designer, *Canadian Nuclear Laboratories* *Sep 2023 – Dec 2023*

- Verified equipment tag status through field walkdowns and database reviews, then updated Engineering Change Control records and **AutoCAD** drawings to improve documentation accuracy.

Microsoft Azure/AI Project member, *University of Waterloo* *May 2022 – Aug 2022*

- Built and trained a machine learning model in **Azure Machine Learning**, earning an “Outstanding” evaluation for project performance.

RELEVANT PROJECTS

Real-Time Operating Systems Project (*C, ARM Cortex-M, ARM Assembly, git*) *May 2025 – Jul 2025*

- Implemented a **cooperative multitasking kernel** with task creation, context switching, and scheduling policies.
- Integrated SVC/PendSV system calls and a custom allocator to support **ARM Cortex-M** task and memory management.

Digital Hardware Systems Project (*SystemVerilog, Vivado, FPGA, git*) *May 2025 – Jul 2025*

- Designed and verified **SystemVerilog** data paths, control FSMs, and FIFO-based modules for FPGA systems.
- Applied pipelining, retiming, and timing analysis to improve throughput and meet setup/hold constraints.

Embedded Microprocessor Systems Project (*C, Verilog, Quartus, Nios II*) *Sep 2024 – Nov 2024*

- Built a **Nios II FPGA-based embedded system** with custom peripherals and hardware/software integration.
- Programmed embedded **C** with polling, interrupts, and register-level access to validate FPGA peripherals.

Instrumentation and Prototyping Lab Project (*Nucleo, STM32Cube, Proteus*) *Sep 2024 – Nov 2024*

- Programmed **STM32 peripherals** including GPIO, PWM, UART, ADC, and timers for sensor/actuator control.
- Designed and assembled a **prototype PCB** through circuit validation, soldering, and end-to-end embedded testing.

EDUCATION

Bachelor of Applied Science in Computer Engineering Expected April 2027

University of Waterloo, Waterloo, Ontario